

 UNIVERSITY OF TECHNOLOGY VNUHCM FACULTY OF APPLIED SCIENCE	MIDTERM EXAM	Semester/Academic year: 241/2024-2025	
		Date: Oct 16	
	Course title	CALCULUS 2	Question sheet code: 2221
	Course ID	MT1005	
	Duration	50 minutes	
• This is a closed book examination. Students are allowed to use standard pocket calculators. • You are required to fill in your full name and student ID on this paper.			
Student's full name: Student ID: Invigilator 1: Invigilator 2:			

- Numeric options (choices) in each question were rounded to the fourth decimal digit.
- There are 16 questions. Each correct answer will earn 0.625 points, while 0.125 points will be deducted for each incorrect answer. Leaving a question blank does not result in a deduction.
- The decimal point . is used when writing decimal numbers, for example: 0.5 (meaning $\frac{1}{2}$), or $1.0x$ (meaning x).
- In each question, select "None of the others" only if the other options are incorrect.

① (L.O.1) A point M has polar coordinates $(r_0, \pi/6)$ with $r_0 > 0$ (in the standard polar coordinate system $x = r \cos \phi, y = r \sin \phi$). Given also that M lies on the circle of equation $(x - 14)^2 + y^2 = 196$, evaluate r_0 .

- A. 24.9487 C. 24.6487 E. None of the others
 B. 24.2487 D. 24.5487

② (L.O.1) Let $f(x, y, z) = 0.9x^2y^3z^4 - 0.3y + 1.0$. Evaluate $f_x(0.9, 0.3, 1.0)$.

- A. None of the others C. 0.4437 E. 0.9437
 B. -0.3563 D. 0.0437

③ (L.O.1) In $\mathbb{R}^3$, which equation from below gives an equation of a paraboloid?

- A. $z = \sqrt{1 + x^2 + y^2}$ D. $z = x^2 + y^2$
 B. None of the others
 C. $z = \sqrt{x^2 + y^2}$ E. $z = -\sqrt{x^2 + y^2}$

④ (L.O.1) How many saddle points does the function $f(x, y) = 2(x - 13)^2 + (y - 8)^2 + 9$ have?

- A. 3 D. 1
 B. None of the others
 C. 2 E. 0

⑤ (L.O.1) Let $f(x, y)$ be a function so that $f_x(-120, 25) = 0.3$ and $f_y(-120, 25) = 0.6$. Consider the function

$$g(t, s) = f(t - st^3, t^2s).$$

Evaluate $g_s(5, 1)$.

- A. -22.2 C. None of the others E. -22.9
 B. -22.5 D. -22.3

6 (L.O.1) Let (P) be the tangent plane to the paraboloid $z = 4x^2 + 6y^2 - 7$ at the point $M(1, 2, 21)$. Which point from the following belongs to (P) ?

- A. None of the others C. $(5, 3, 72)$ E. $(5, 3, 77)$
 B. $(1, 2, 26)$ D. $(1, 2, 23)$

7 (L.O.1) Evaluate the double integral

$$\iint_D x^2 dA,$$

where D is the region that lies to the right of the y -axis within the circle $x^2 + y^2 = 3^2$.

- A. None of the others C. 32.1086 E. 31.8086
 B. 31.5086 D. 32.7086

8 (L.O.1) Evaluate the double integral

$$\iint_D 8x^2 dA,$$

where D is the region bounded by the curve $y = 12 - x^2$ and the line $y = 0$.

- A. None of the others C. 1063.972 E. 1064.372
 B. 1063.772 D. 1064.172

9 (L.O.1) Which function whose domain is enclosed by a disk?

- A. $f(x, y) = \ln(1 + x^2 + y^2)$ D. $f(x, y) = \cosh(1 - x^2 - y^2)$
 B. $f(x, y) = \sin(x^2 - y^3)$
 C. None of the others E. $f(x, y) = 1 - x^2 - y^2$

10 (L.O.1) Let $z(x, y)$ be a function defined implicitly from the equation

$$0.6xz^2 = z^5 + 1.1xz + y.$$

Evaluate $z_x(0, 1)$.

- A. 0.34 C. 0.84 E. 1.24
 B. None of the others D. 0.04

11 (L.O.2) Let D be a lamina bounded by the triangle with vertices $(0, 0)$, $(13, 0)$ and $(0, 13)$. Evaluate the mass of the lamina if the density function is $\rho(x, y) = xy$.

- | | | |
|--------------|-----------------------|--------------|
| A. 1190.9417 | C. 1190.0417 | E. 1190.7417 |
| B. 1190.4417 | D. None of the others | |

- 12 (L.O.2) A production line is utilizing two different types of industrial machines for its operation. The quantity of products produced per day by the production line is determined by:

$$Q = 300xy^2 \text{ (units),}$$

where x and y represent the operating hours of these two types of machines, respectively. Assume that the total operating hours per day for both types of machines must not exceed 16 hours. Calculate the maximum quantity of products (in units of products) that the production line can generate in a day.

- | | | |
|----------------|-----------------------|----------------|
| A. 182048.9444 | C. None of the others | E. 182044.4444 |
| B. 182035.4444 | D. 182050.3444 | |

- 13 (L.O.1) Let $\mathbf{n} = \langle a, b, c \rangle$ be the unit normal vector to the surface $z = 5 - 6x^2 - 7y^2$ at the point $M(1, -2, -29)$ in which $\mathbf{n}$ is pointing upward (namely $\widehat{\mathbf{n}, \overrightarrow{Oz}} < 90^\circ$). Find the value of c .

- | | | |
|-----------------------|-----------|-----------|
| A. None of the others | C. 0.2328 | E. 0.5328 |
| B. -0.3672 | D. 0.0328 | |

- 14 (L.O.1) Let $f(x, y)$ be a function such that $f(2, 5) = 0.8$, $f_x(2, 5) = -0.3$, and $f_y(2, 5) = 1.2$. Use the linear approximation to obtain an approximation for $f(2.01, 4.98)$.

- | | | |
|-----------------------|----------|----------|
| A. 0.3865 | C. 0.827 | E. 0.773 |
| B. None of the others | D. 0.821 | |

- 15 (L.O.2) Find the minimum value of the function $f(x, y) = 1.1 + 2.6(x - 0.9)^2 + 3.4(y - 1.7)^2$ on the disk $x^2 + y^2 \leq 4.7$.

- | | | |
|--------|-----------------------|--------|
| A. 0.7 | C. 0.8 | E. 1.5 |
| B. 1.1 | D. None of the others | |

- 16 (L.O.1) How many critical points does the function $f(x, y) = 13x^2 - 14y + 9$ have?

- | | | |
|-----------------------|------|------|
| A. None of the others | C. 1 | E. 0 |
| B. 2 | D. 3 | |

END

1. (B)
2. (D)
3. (D)
4. (E)
5. (B)
6. (E)
7. (E)
8. (D)
9. (C)
10. (A)
11. (C)
12. (E)
13. (D)
14. (E)
15. (B)
16. (E)

 UNIVERSITY OF TECHNOLOGY VNUHCM FACULTY OF APPLIED SCIENCE	MIDTERM EXAM	Semester/Academic year: 241/2024-2025		
		Date: Oct 16		
	Course title	CALCULUS 2	Question sheet code: 2222	
	Course ID	MT1005		
	Duration	50 minutes		
• This is a closed book examination. Students are allowed to use standard pocket calculators. • You are required to fill in your full name and student ID on this paper.				
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- The decimal point . is used when writing decimal numbers, for example: 0.5 (meaning $\frac{1}{2}$), or $1.0x$ (meaning x).
- In each question, select "None of the others" only if the other options are incorrect.

① (L.O.1) Evaluate the double integral

$$\iint_D x^2 dA,$$

where D is the region that lies to the right of the y -axis within the circle $x^2 + y^2 = 3^2$.

- A. 31.5086 C. 32.7086 E. 31.8086
 B. 32.1086 D. None of the others

② (L.O.1) Let $\mathbf{n} = \langle a, b, c \rangle$ be the unit normal vector to the surface $z = 5 - 6x^2 - 7y^2$ at the point $M(1, -2, -29)$ in which $\mathbf{n}$ is pointing upward (namely $\widehat{\mathbf{n}, \vec{Oz}} < 90^\circ$). Find the value of c .

- A. 0.2328 C. 0.0328 E. 0.5328
 B. -0.3672 D. None of the others

③ (L.O.2) A production line is utilizing two different types of industrial machines for its operation. The quantity of products produced per day by the production line is determined by:

$$Q = 300xy^2 \text{ (units),}$$

where x and y represent the operating hours of these two types of machines, respectively. Assume that the total operating hours per day for both types of machines must not exceed 16 hours. Calculate the maximum quantity of products (in units of products) that the production line can generate in a day.

- A. 182035.4444
B. 182044.4444

- C. 182050.3444
D. 182048.9444

E. None of the others

- ④ (L.O.1) Let $f(x, y)$ be a function so that $f_x(-120, 25) = 0.3$ and $f_y(-120, 25) = 0.6$. Consider the function

$$g(t, s) = f(t - st^3, t^2s).$$

Evaluate $g_s(5, 1)$.

- A. -22.3
B. -22.2

- C. None of the others
D. -22.5

E. -22.9

- ⑤ (L.O.1) How many saddle points does the function $f(x, y) = 2(x - 13)^2 + (y - 8)^2 + 9$ have?

- A. None of the others
B. 3
C. 2

- D. 1
E. 0

- ⑥ (L.O.1) A point M has polar coordinates $(r_0, \pi/6)$ with $r_0 > 0$ (in the standard polar coordinate system $x = r \cos \phi, y = r \sin \phi$). Given also that M lies on the circle of equation $(x - 14)^2 + y^2 = 196$, evaluate r_0 .

- A. 24.6487
B. None of the others

- C. 24.5487
D. 24.2487

E. 24.9487

- ⑦ (L.O.1) In $\mathbb{R}^3$, which equation from below gives an equation of a paraboloid?

- A. $z = \sqrt{1 + x^2 + y^2}$
B. None of the others
C. $z = -\sqrt{x^2 + y^2}$

- D. $z = \sqrt{x^2 + y^2}$
E. $z = x^2 + y^2$

- ⑧ (L.O.2) Let D be a lamina bounded by the triangle with vertices $(0, 0)$, $(13, 0)$ and $(0, 13)$. Evaluate the mass of the lamina if the density function is $\rho(x, y) = xy$.

- A. 1190.9417
B. None of the others

- C. 1190.0417
D. 1190.4417

E. 1190.7417

- ⑨ (L.O.1) Let $z(x, y)$ be a function defined implicitly from the equation

$$0.6xz^2 = z^5 + 1.1xz + y.$$

Evaluate $z_x(0, 1)$.

- A. None of the others
B. 0.04

- C. 1.24
D. 0.84

E. 0.34

- ⑩ (L.O.1) Let $f(x, y)$ be a function such that $f(2, 5) = 0.8$, $f_x(2, 5) = -0.3$, and $f_y(2, 5) = 1.2$. Use the linear approximation to obtain an approximation for $f(2.01, 4.98)$.

- A. 0.773 C. 0.827 E. None of the others
 B. 0.3865 D. 0.821

11 (L.O.1) Which function whose domain is enclosed by a disk?

- A. $f(x, y) = \ln(1 + x^2 + y^2)$ D. None of the others
 B. $f(x, y) = \sin(x^2 - y^3)$
 C. $f(x, y) = \cosh(1 - x^2 - y^2)$ E. $f(x, y) = 1 - x^2 - y^2$

12 (L.O.1) Evaluate the double integral

$$\iint_D 8x^2 dA,$$

where D is the region bounded by the curve $y = 12 - x^2$ and the line $y = 0$.

- A. 1063.772 C. None of the others E. 1064.172
 B. 1064.372 D. 1063.972

13 (L.O.1) Let (P) be the tangent plane to the paraboloid $z = 4x^2 + 6y^2 - 7$ at the point $M(1, 2, 21)$. Which point from the following belongs to (P) ?

- A. $(1, 2, 23)$ C. $(5, 3, 72)$ E. $(5, 3, 77)$
 B. $(1, 2, 26)$ D. None of the others

14 (L.O.2) Find the minimum value of the function $f(x, y) = 1.1 + 2.6(x - 0.9)^2 + 3.4(y - 1.7)^2$ on the disk $x^2 + y^2 \leq 4.7$.

- A. None of the others C. 0.8 E. 0.7
 B. 1.1 D. 1.5

15 (L.O.1) How many critical points does the function $f(x, y) = 13x^2 - 14y + 9$ have?

- A. None of the others C. 3 E. 1
 B. 2 D. 0

16 (L.O.1) Let $f(x, y, z) = 0.9x^2y^3z^4 - 0.3y + 1.0$. Evaluate $f_x(0.9, 0.3, 1.0)$.

- A. 0.9437 C. 0.0437 E. -0.3563
 B. 0.4437 D. None of the others

END

1. (E)
2. (C)
3. (B)
4. (D)
5. (E)
6. (D)
7. (E)
8. (C)
9. (E)
10. (A)
11. (D)
12. (E)
13. (E)
14. (B)
15. (D)
16. (C)

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- A. 24.2487 C. 24.6487 E. 24.5487
 B. None of the others D. 24.9487

③ (L.O.1) Evaluate the double integral

$$\iint_D x^2 dA,$$

where D is the region that lies to the right of the y -axis within the circle $x^2 + y^2 = 3^2$.

- A. 31.5086 C. 32.7086 E. 31.8086
 B. None of the others D. 32.1086

④ (L.O.1) In $\mathbb{R}^3$, which equation from below gives an equation of a paraboloid?

A. $z = \sqrt{x^2 + y^2}$

D. $z = \sqrt{1 + x^2 + y^2}$

B. $z = x^2 + y^2$

C. None of the others

E. $z = -\sqrt{x^2 + y^2}$

- 5 (L.O.2) A production line is utilizing two different types of industrial machines for its operation. The quantity of products produced per day by the production line is determined by:

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A. 182048.9444

C. 182035.4444

E. None of the others

B. 182050.3444

D. 182044.4444

- 6 (L.O.1) Let $z(x, y)$ be a function defined implicitly from the equation

$$0.6xz^2 = z^5 + 1.1xz + y.$$

Evaluate $z_x(0, 1)$.

A. 0.04

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E. None of the others

B. 0.84

D. 1.24

- 7 (L.O.1) Which function whose domain is enclosed by a disk?

A. $f(x, y) = \ln(1 + x^2 + y^2)$

D. $f(x, y) = \cosh(1 - x^2 - y^2)$

B. None of the others

C. $f(x, y) = \sin(x^2 - y^3)$

E. $f(x, y) = 1 - x^2 - y^2$

- 8 (L.O.1) Let $f(x, y, z) = 0.9x^2y^3z^4 - 0.3y + 1.0$. Evaluate $f_x(0.9, 0.3, 1.0)$.

A. 0.4437

C. None of the others

E. -0.3563

B. 0.9437

D. 0.0437

- 9 (L.O.1) Let $f(x, y)$ be a function so that $f_x(-120, 25) = 0.3$ and $f_y(-120, 25) = 0.6$. Consider the function

$$g(t, s) = f(t - st^3, t^2s).$$

Evaluate $g_s(5, 1)$.

A. None of the others

C. -22.3

E. -22.2

B. -22.9

D. -22.5

- 10 (L.O.1) How many critical points does the function $f(x, y) = 13x^2 - 14y + 9$ have?

- A. 3 C. 0 E. None of the others
B. 2 D. 1

- 11** (L.O.2) Find the minimum value of the function $f(x, y) = 1.1 + 2.6(x - 0.9)^2 + 3.4(y - 1.7)^2$ on the disk $x^2 + y^2 \leq 4.7$.

- A. 0.8 C. 1.5 E. None of the others
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- 12** (L.O.2) Let D be a lamina bounded by the triangle with vertices $(0,0)$, $(13,0)$ and $(0,13)$. Evaluate the mass of the lamina if the density function is $\rho(x,y) = xy$.

- A. 1190.0417 C. 1190.4417 E. 1190.9417
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- A. -0.3672 C. 0.5328 E. 0.0328
B. None of the others D. 0.2328

- 15** (L.O.1) How many saddle points does the function $f(x, y) = 2(x - 13)^2 + (y - 8)^2 + 9$ have?

- A. 1
B. 2
C. None of the others
- D. 0
E. 3

- 16** (L.O.1) Let $f(x, y)$ be a function such that $f(2, 5) = 0.8$, $f_x(2, 5) = -0.3$, and $f_y(2, 5) = 1.2$. Use the linear approximation to obtain an approximation for $f(2.01, 4.98)$.

- A. 0.773 C. None of the others E. 0.827
B. 0.821 D. 0.3865

-END

1. (D)
2. (A)
3. (E)
4. (B)
5. (D)
6. (C)
7. (B)
8. (D)
9. (D)
10. (C)
11. (D)
12. (A)
13. (A)
14. (E)
15. (D)
16. (A)

 UNIVERSITY OF TECHNOLOGY VNUHCM FACULTY OF APPLIED SCIENCE	MIDTERM EXAM	Semester/Academic year: 241/2024-2025	
		Date: Oct 16	
	Course title	CALCULUS 2	Question sheet code: 2224
	Course ID	MT1005	
	Duration	50 minutes	
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Evaluate $z_x(0, 1)$.

- A. 0.84 C. None of the others E. 0.04
 B. 1.24 D. 0.34

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- A. 3 D. 0
 B. 2
 C. None of the others E. 1

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- A. 1.1 C. 0.8 E. None of the others
 B. 0.7 D. 1.5
- 13 (L.O.2) Let D be a lamina bounded by the triangle with vertices $(0, 0)$, $(13, 0)$ and $(0, 13)$. Evaluate the mass of the lamina if the density function is $\rho(x, y) = xy$.
- A. 1190.9417 C. None of the others E. 1190.4417
 B. 1190.0417 D. 1190.7417
- 14 (L.O.1) How many critical points does the function $f(x, y) = 13x^2 - 14y + 9$ have?
- A. 3 C. 2 E. None of the others
 B. 0 D. 1
- 15 (L.O.2) A production line is utilizing two different types of industrial machines for its operation. The quantity of products produced per day by the production line is determined by:
- $$Q = 300xy^2 \text{ (units),}$$
- where x and y represent the operating hours of these two types of machines, respectively. Assume that the total operating hours per day for both types of machines must not exceed 16 hours. Calculate the maximum quantity of products (in units of products) that the production line can generate in a day.
- A. 182035.4444 C. 182044.4444 E. 182048.9444
 B. None of the others D. 182050.3444
- 16 (L.O.1) In $\mathbb{R}^3$, which equation from below gives an equation of a paraboloid?
- A. $z = -\sqrt{x^2 + y^2}$ D. $z = \sqrt{1 + x^2 + y^2}$
 B. $z = x^2 + y^2$
 C. None of the others E. $z = \sqrt{x^2 + y^2}$

END

1. (E)
2. (E)
3. (C)
4. (A)
5. (E)
6. (D)
7. (D)
8. (B)
9. (D)
10. (E)
11. (A)
12. (A)
13. (B)
14. (B)
15. (C)
16. (B)